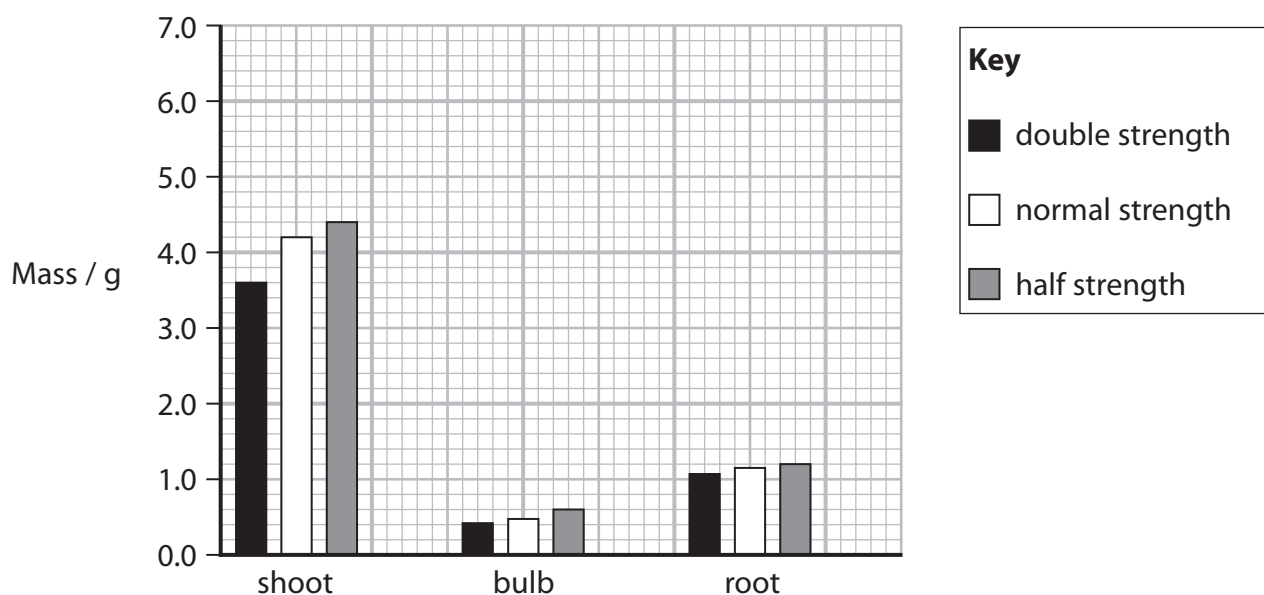


(ii) The graph shows the results of this investigation.



The shoot and the bulb of an onion plant are edible.

The table shows the percentage of the total onion plant that is edible.

Percentage of onion plant that is edible (%)		
when grown in double strength	when grown in normal strength	when grown in half strength
79.0	80.2	

Complete the table by calculating the percentage of the onion plant mass that is edible in the onions grown in the half strength solution.

(2)



- (iii) Describe **three** conclusions that can be drawn from the results of this investigation.

Use the information in the graph and table to support your answer.

(3)

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2

3

- (iv) Describe the information that is needed to decide if any of the differences seen are statistically significant.

(3)

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(b) The table shows the mineral salts used to make a complete mineral ion solution.

Mineral salt formula	Mineral salt name
KNO_3	potassium nitrate
$\text{Ca}(\text{NO}_3)_2$	calcium nitrate
MgSO_4	magnesium sulfate
KH_2PO_4	potassium hydrogen phosphate

The effect of mineral ion deficiencies can be investigated by growing onion plants in solutions that have one mineral ion missing.

Suggest how a complete mineral ion solution could be modified to show the effect of phosphate ions on the growth of onion plants.

(2)

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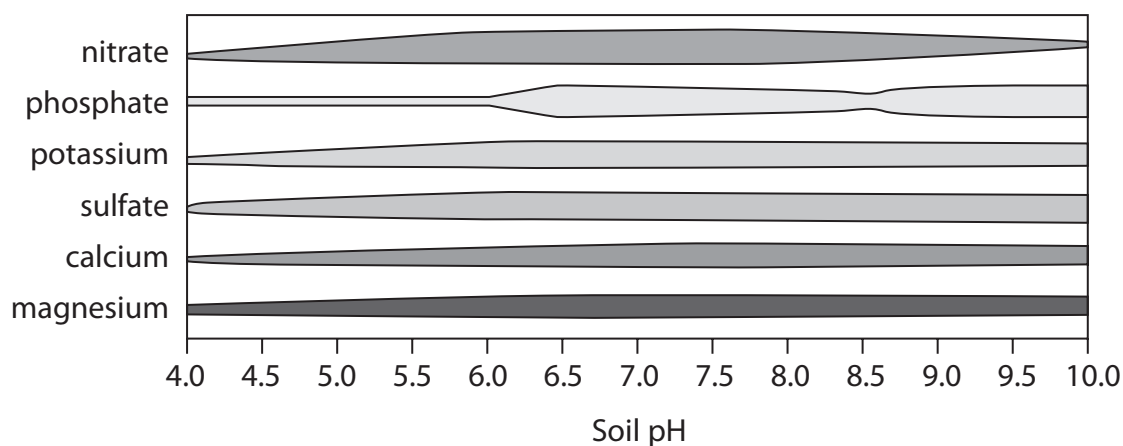
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(c) The pH of the soil can affect the uptake of mineral ions by plants.

The chart shows the effect of pH on the uptake of some mineral ions.

The height of the bar for each mineral ion indicates the uptake of that mineral ion.



The optimum pH for the growth of these plants is between 6.5 and 8.0.

Explain why the optimum pH is within this range.

Use the information in the chart to support your answer.

(4)

(Total for Question 3 = 18 marks)

TOTAL FOR PAPER = 50 MARKS